Roll No. | | | | | | | |

Mid Semester Examination 2025

Name of the Program: **B. Tech. Biotechnology** Semester: **V**

Name of the Course: **Nanobiotechnology** Course Code: **TBT-506**

Time: **90** minutes Maximum Marks: **50**

Note:-

(i) Answer all the questions by choosing any one of the sub-questions

(ii) Each question carries 10 marks

Q1	(10 marks)	CO1
(a)	What are carbon nanotubes (**CNTs**)? Discuss their types, basis of classification, and properties (mechanical, electrical, and thermal).	
OR		
(b)	Briefly explain **CVD** and **Laser Ablation** methods of synthesizing CNTs.	
Q2	(10 marks)	CO1
(a)	Give an account of the discovery of CNTs and explain Arc Discharge method of synthesizing CNT.	
OR		
(b)	What is significance of nano-domain? Briefly discuss about different forces involved at nano-domain.	
Q3	(10 marks)	CO1 & CO2
(a)	Define quantum dots, and give a detailed account of their application as bio-imaging probe.	
OR		
(b)	Discuss basic structure of a virus. With help of appropriate diagrams, describe how **M13 bacteriophage** and **Tobacco mosaic virus (TMV)** can be used for nanoparticle synthesis?	
Q4	(10 marks)	CO2
(a)	What do you understand by green synthesis? How nanoparticles can be synthesized using plant metabolites and microbes?	
OR		
(b)	Briefly describe **Sol-Gel Technique** of fabricating nanomaterials along with its general mechanism, reaction characteristics and advantages.	
Q5	(10 marks)	CO2
(a)	Discuss **co-precipitation** method of nanoparticle fabrication and **Ostwald Ripening** process.	
OR		
(b)	What are **microemulsions**? Discuss their fabrication methods, advantages, and disadvantages.	